A. Gregor and Cryptography

time limit per test 1 second
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 800 | Tags: math, number theory | https://codeforces.com/problemset/problem/1549/A

Gregor is learning about RSA cryptography, and although he doesn't understand how RSA works, he is now fascinated with prime numbers and factoring them.

Gregor's favorite **prime** number is P. Gregor wants to find two *bases* of P. Formally, Gregor is looking for two integers a and b which satisfy both of the following properties.

- $P \bmod a = P \bmod b$, where $x \bmod y$ denotes the remainder when x is divided by y, and
- $2 \le a \text{ lt } b \le P$.

Help Gregor find two bases of his favorite prime number!

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \le t \le 1000$).

Each subsequent line contains the integer P ($5 \le P \le 10^9$), with P guaranteed to be prime.

Output

Your output should consist of t lines. Each line should consist of two integers a and b ($2 \le a \text{ lt } b \le P$). If there are multiple possible solutions, print any.

Example

Input
2 17 5

Output
3 5 2 4

Note

The first query is P=17. a=3 and b=5 are valid *bases* in this case, because 17 mod 3 = 17 mod 5 = 2. There are other pairs which work as well.

In the second query, with P=5, the only solution is a=2 and b=4.